

Aether vs Alternatives

Why a Universal Runtime Control Plane?

Compare Aether to kubectl/Helm, HashiCorp Nomad, Docker Compose, and custom scripts. One spec, four runtimes, zero lock-in.

Podman • Kubernetes • KubeVirt • Metal3

Head-to-Head Comparison

How Aether stacks up against the most common deployment tools across key dimensions.

Dimension	Aether	kubectl / Helm	Nomad	Docker Compose	Custom Scripts
Runtimes	4 (Podman, K8s, KubeVirt, Metal3)	1 (K8s only)	3 (Docker, exec, QEMU)	1 (Docker only)	Varies
Spec Format	One unified YAML	K8s manifests + Helm charts	HCL job files	docker-compose.yml	Shell/Python/ad-hoc
Migration Engine	16 paths, 3 strategies	None	None	None	Manual
Decision Engine	Auto-selects runtime	User picks K8s	Manual scheduler hints	Always Docker	Hardcoded
Drift Detection	Built-in + auto-reconcile	Partial (flux/argo)	Partial (plan diff)	None	None
Secrets Management	AES-256-GCM built-in	External (Vault/SOPS)	Built-in Vault integration	env files	env files
SLA Monitoring	Built-in with error budgets	External tools	External tools	None	None

Language	Rust (single binary)	Go	Go	Go	Varies

Unique Advantages

Aether is not just another deployment tool — it is the only control plane that unifies four runtimes behind a single declarative spec.



One Spec, Four Runtimes

Write your workload YAML once. Aether deploys it to Podman for dev, Kubernetes for staging, KubeVirt for legacy apps, and Metal3 for GPU workloads — without rewriting a single line.



Migration Engine

16 distinct runtime-to-runtime migration paths with blue-green, rolling, and immediate strategies. Move workloads between any two runtimes with zero downtime and automatic rollback.



Decision Engine

The built-in scheduler evaluates CPU, memory, GPU, cost, and affinity to automatically select the optimal runtime. Override when you need to, trust it when you do not.



Full Observability Stack

Drift detection, SLA compliance, metrics, health monitoring, and audit trails — all built into the control plane. No external Prometheus, Grafana, or Vault required to get started.

Other tools manage one runtime well. Aether manages the choice between runtimes — and makes that choice reversible at any time.

1

Spec for All Runtimes

16

Migration Paths

0

External Dependencies

Feature Matrix

A comprehensive comparison of capabilities across Aether and competing tools.

Feature	Aether	kubectl/Helm	Nomad	Compose	Scripts
Container deployment	✓	✓	✓	✓	✓
VM deployment (KubeVirt)	✓	✗	✓	✗	✗
Bare-metal provisioning	✓	✗	✗	✗	✗
Cross-runtime migration	✓	✗	✗	✗	✗
Auto runtime selection	✓	✗	✗	✗	✗
Built-in secrets (AES-256)	✓	✗	✓	✗	✗
Drift detection	✓	✗	✗	✗	✗
SLA error budgets	✓	✗	✗	✗	✗
Environment promotion	✓	✗	✗	✗	✗
Policy engine	✓	✓	✓	✗	✗
Tamper-evident audit trail	✓	✗	✓	✗	✗
Single binary, no daemon	✓	✓	✗	✗	✓

Migration from Existing Tools

Aether is designed to complement, not replace. Adopt incrementally alongside your existing tooling.

Migration Paths

1

From Docker Compose

Import your `docker-compose.yml` using `aether compose import`. Aether converts services to workload specs, preserving port mappings, environment variables, and volume mounts. Run on Podman first, then migrate to Kubernetes when ready.

2

From kubectl / Helm

Start by managing new workloads with Aether specs while keeping existing Helm releases. Gradually migrate services by creating Aether specs and using the migration engine to move from Kubernetes to any other runtime (or keep on Kubernetes with unified management).

3

From Nomad

Translate HCL job files to Aether YAML specs. Aether's scheduler provides similar constraint-based placement but adds cross-runtime migration and built-in drift detection that Nomad lacks.

4

From Custom Scripts

Replace shell scripts with declarative YAML specs. Gain idempotent deploys, automatic rollback, state tracking, and audit trails. The initial conversion is the last time you write deployment logic.

When to Use Aether



Ideal For

- Teams using multiple runtimes (containers + VMs + bare-metal)
- Organizations needing cross-runtime migration
- Teams wanting a single unified control plane



Consider Alternatives When

- You only use Kubernetes and always will
- You need a service mesh (Istio/Linkerd)
- You require a full-featured secrets vault with HSM

- Environments requiring built-in secrets and SLA monitoring
- Edge deployments needing lightweight, single-binary tooling

- Your team has deep Helm/Nomad expertise and no multi-runtime needs
- You need cloud provider managed services (ECS, Cloud Run)

Choose **Aether**

The only control plane that treats containers, VMs, and bare-metal as first-class citizens.

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Runtimes Unified

40+

CLI Commands

1

Single Binary

Start Deploying Across Every Runtime Today

Aether is open source, written in Rust, and designed for production. One spec, four runtimes, zero lock-in. Install in seconds, deploy in minutes.

```
# Get started in 3 commands
$ cargo install aether
$ aether init my-project
$ aether run --spec workload.yaml

# Migrate from Docker Compose
$ aether compose import docker-compose.yml
$ aether run --spec imported-workload.yaml

# Move to Kubernetes when ready
$ aether migrate my-app --from podman --to kubernetes \
  --strategy blue-green
```